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Polymarket Bias & Mispricing Analysis

Closed Markets 2025 — Calibration, Drift, Breakouts & Exploitable Patterns (v3)

Laurence

2026-02-18

1 · Setup & Data Loading

Show

Show

```
## 111178 market rows loaded
```

Show

```
## 87 columns
```

1.1 Parse Resolutions & Token Mapping

Show

```
## 49753 binary Yes/No markets with clean resolution
```

Show

```
## Resolution rate (Yes = 1): 0.267
```

1.2 Derive Categories from Tags

Show

```
##
## ── Category distribution ──
```

Show

```
## # A tibble: 8 × 2
##   category          n
##   <chr>         <int>
## 1 Sports        14881
## 2 Crypto        14161
## 3 Politics       9844
## 4 Science/Tech   5570
## 5 Other          2410
## 6 Entertainment  2091
## 7 Economics       777
## 8 World Events     19
```

Show

```
##
## Politics: election-tagged = 4628 / 9844
```

2 · Calibration Analysis (Snapshot Prices)

The core question: **when the market prices a Yes token at probability p, does the event actually happen p% of the time?**

Show

```
## 49753 unique Yes tokens to look up
```

Show

```
## 100 batches of up to 500 tokens
```

Show

```
##   Batch 5/100 done (11595 snapshot rows so far)
##   Batch 10/100 done (23809 snapshot rows so far)
##   Batch 15/100 done (36470 snapshot rows so far)
##   Batch 20/100 done (48962 snapshot rows so far)
##   Batch 25/100 done (61591 snapshot rows so far)
##   Batch 30/100 done (74504 snapshot rows so far)
##   Batch 35/100 done (87249 snapshot rows so far)
##   Batch 40/100 done (100113 snapshot rows so far)
##   Batch 45/100 done (112551 snapshot rows so far)
##   Batch 50/100 done (123420 snapshot rows so far)
##   Batch 55/100 done (133622 snapshot rows so far)
##   Batch 60/100 done (145422 snapshot rows so far)
##   Batch 65/100 done (157735 snapshot rows so far)
##   Batch 70/100 done (169985 snapshot rows so far)
##   Batch 75/100 done (181988 snapshot rows so far)
##   Batch 80/100 done (194828 snapshot rows so far)
##   Batch 85/100 done (207703 snapshot rows so far)
##   Batch 90/100 done (220378 snapshot rows so far)
##   Batch 95/100 done (232904 snapshot rows so far)
##   Batch 100/100 done (245827 snapshot rows so far)
```

Show

```
##            used  (Mb) gc trigger  (Mb)  max used  (Mb)
## Ncells  3526446 188.4    6073394 324.4   6073394 324.4
## Vcells 43049468 328.5  121387422 926.2 121386354 926.2
```

Show

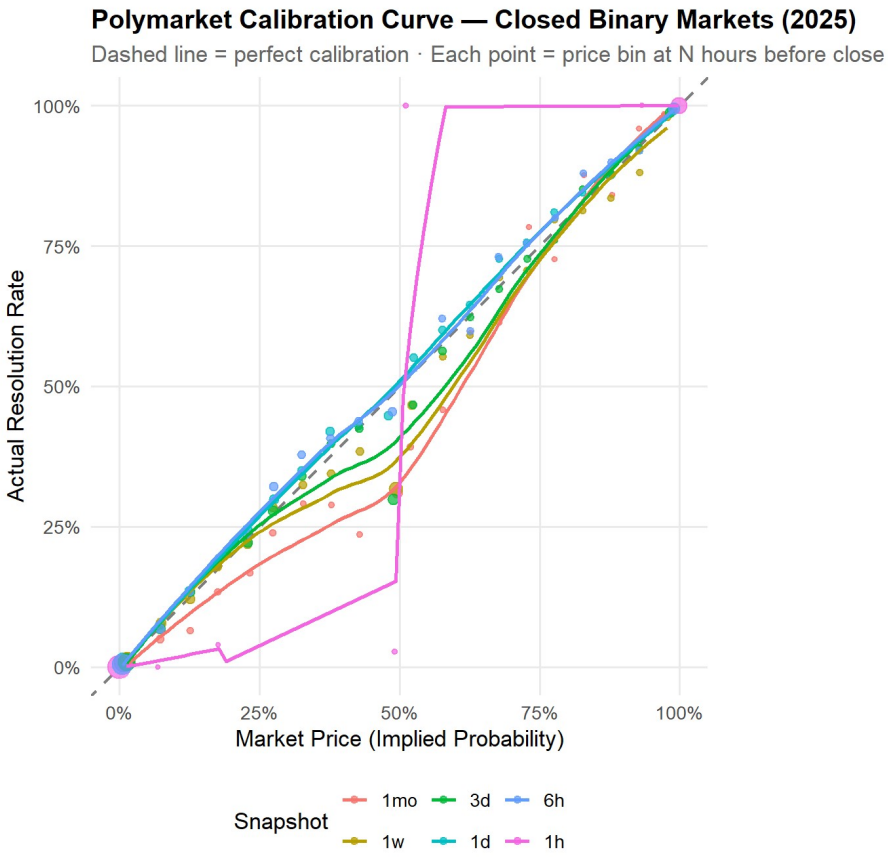
```
##
##  245827 total market × snapshot observations
```

Show

```
## 49753 markets with full price history stats
```

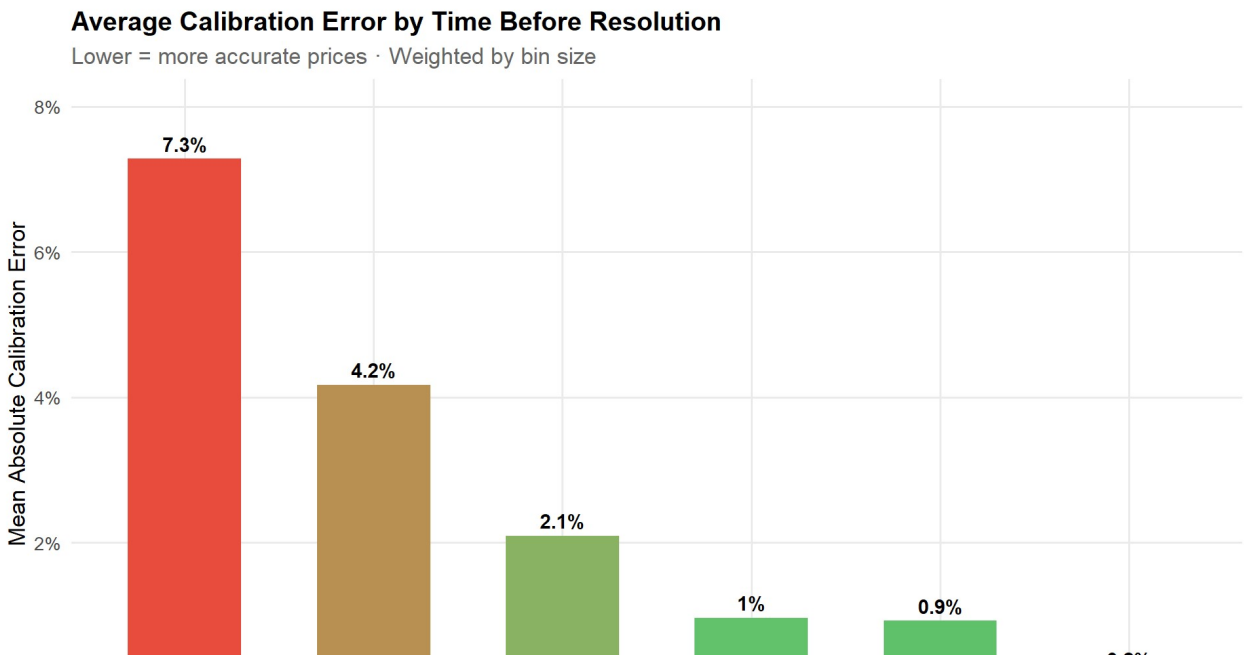
2.1 Overall Calibration Curve

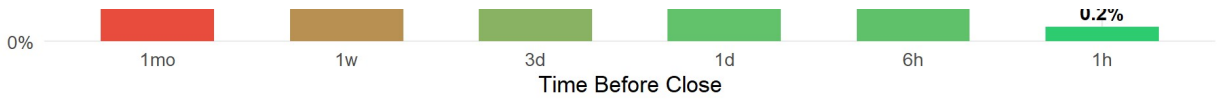
Show



2.2 Calibration Error by Snapshot Horizon

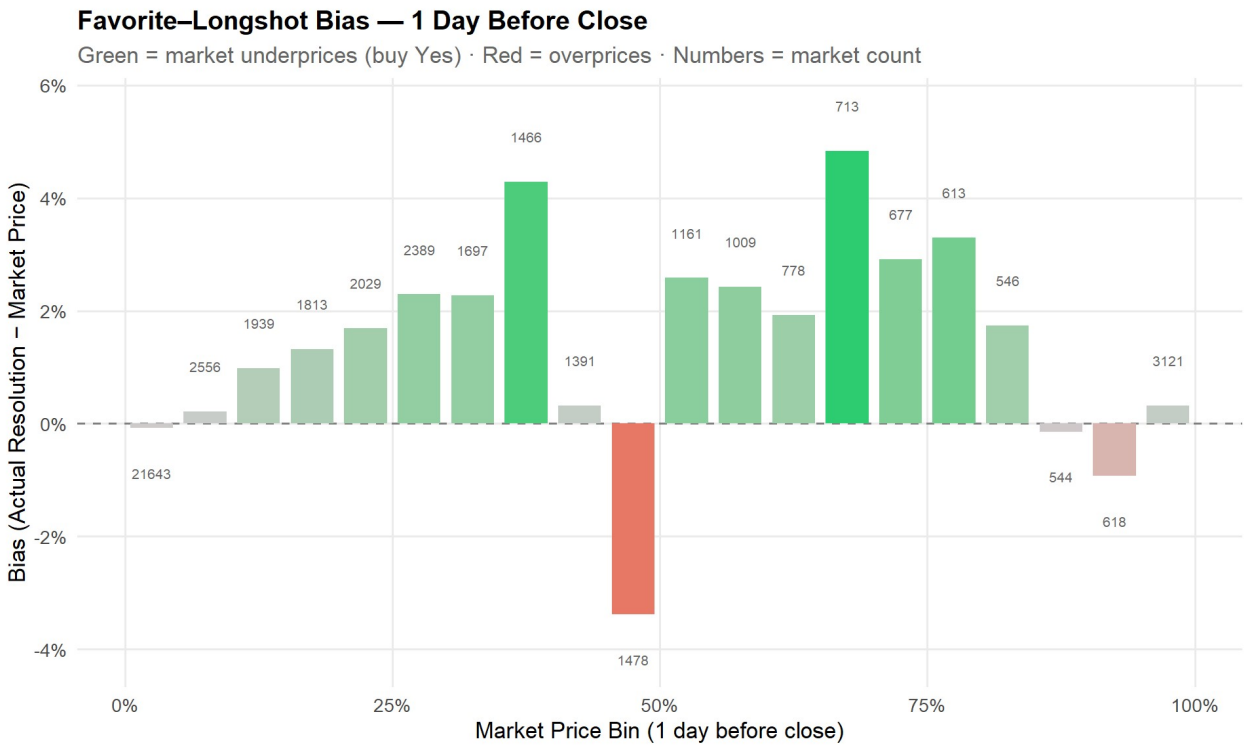
Show





2.3 Favorite–Longshot Bias

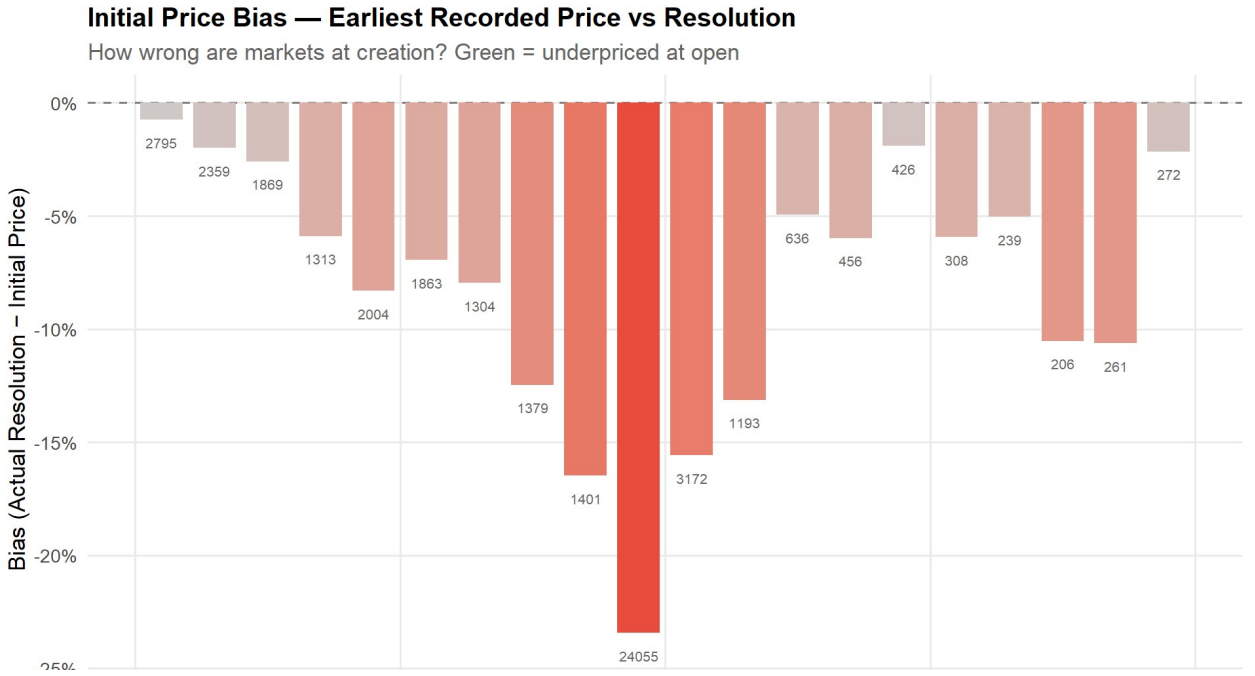
Show

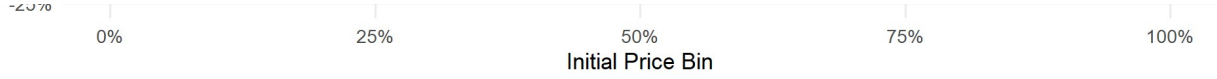


2.4 Initial Price Bias

New. We compare the *earliest* recorded price per market to its resolution. This tests whether market creators systematically anchor opening prices too high or too low — a bias that is separate from, and often larger than, near-close mispricing.

Show





Show

```
## ── Initial vs 1-day-before-close bias comparison ──
```

Show

```
##   At market open:      -15.65%
```

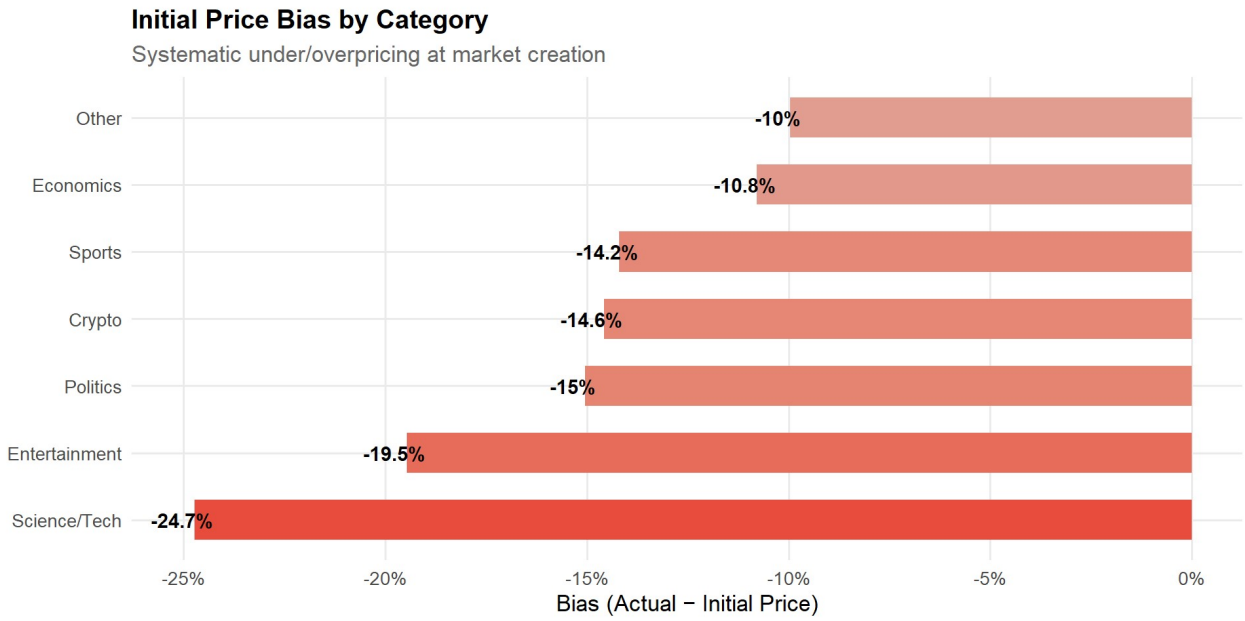
Show

```
##   1 day before close:  +0.75%
```

Show

```
##   Markets compared:    47508
```

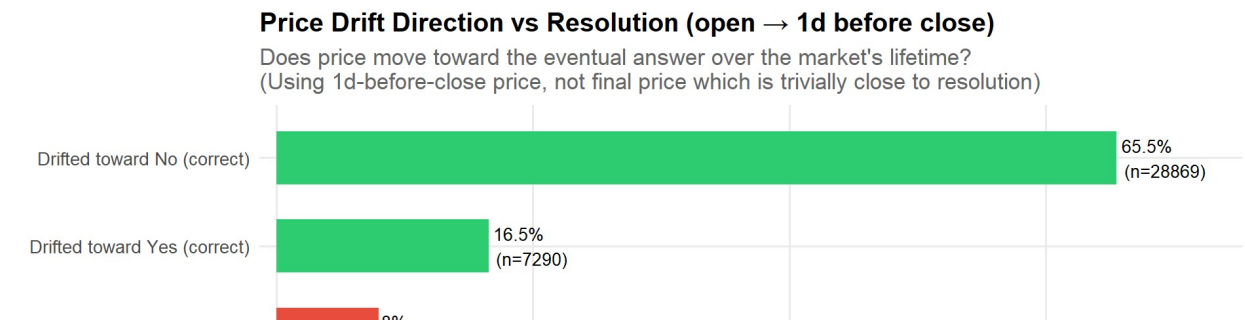
Show

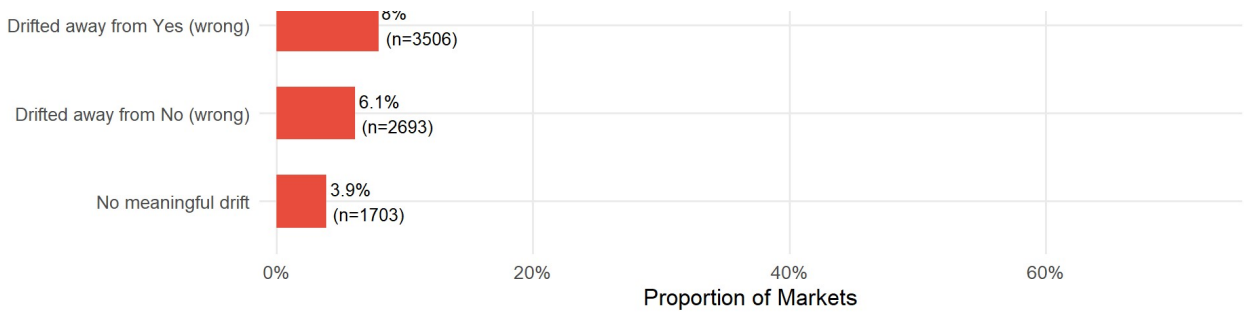


2.5 Price Drift Over Market Lifetime

New. How do prices evolve from open to close? Do they drift monotonically toward resolution, or overshoot and correct?

Show





Show

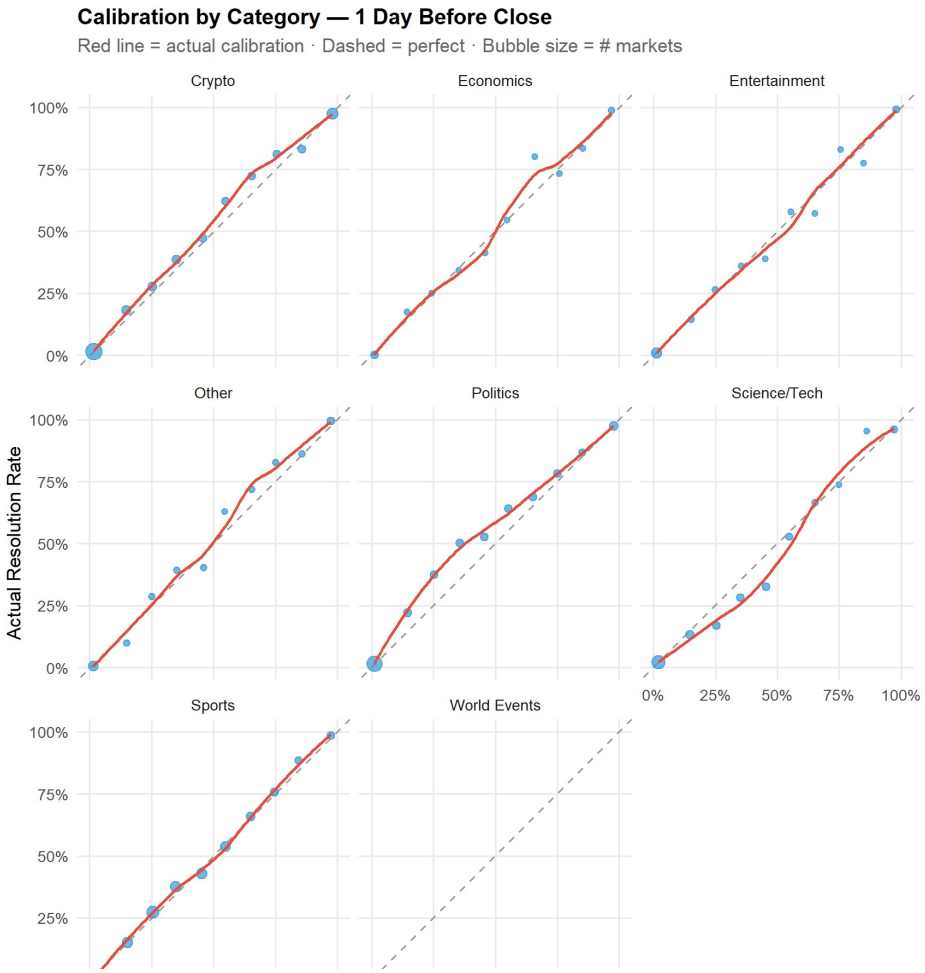
Calibration Quality by Price Range (market volatility)

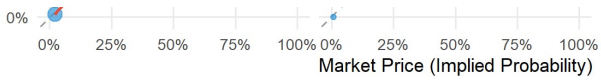
range_tier	n	resolution	init_bias	final_bias	brier_init
Low (<10pp)	2795	0.0497	-0.0378	-0.0053	0.0024
Medium (10-30pp)	5016	0.0889	-0.1321	-0.0342	0.0327
High (30-60pp)	26520	0.1592	-0.2989	-0.0533	0.2061
Very High (>60pp)	13177	0.5653	0.0955	0.1483	0.2993

3 · Category-Level Mispricings

3.1 Category Calibration Curves

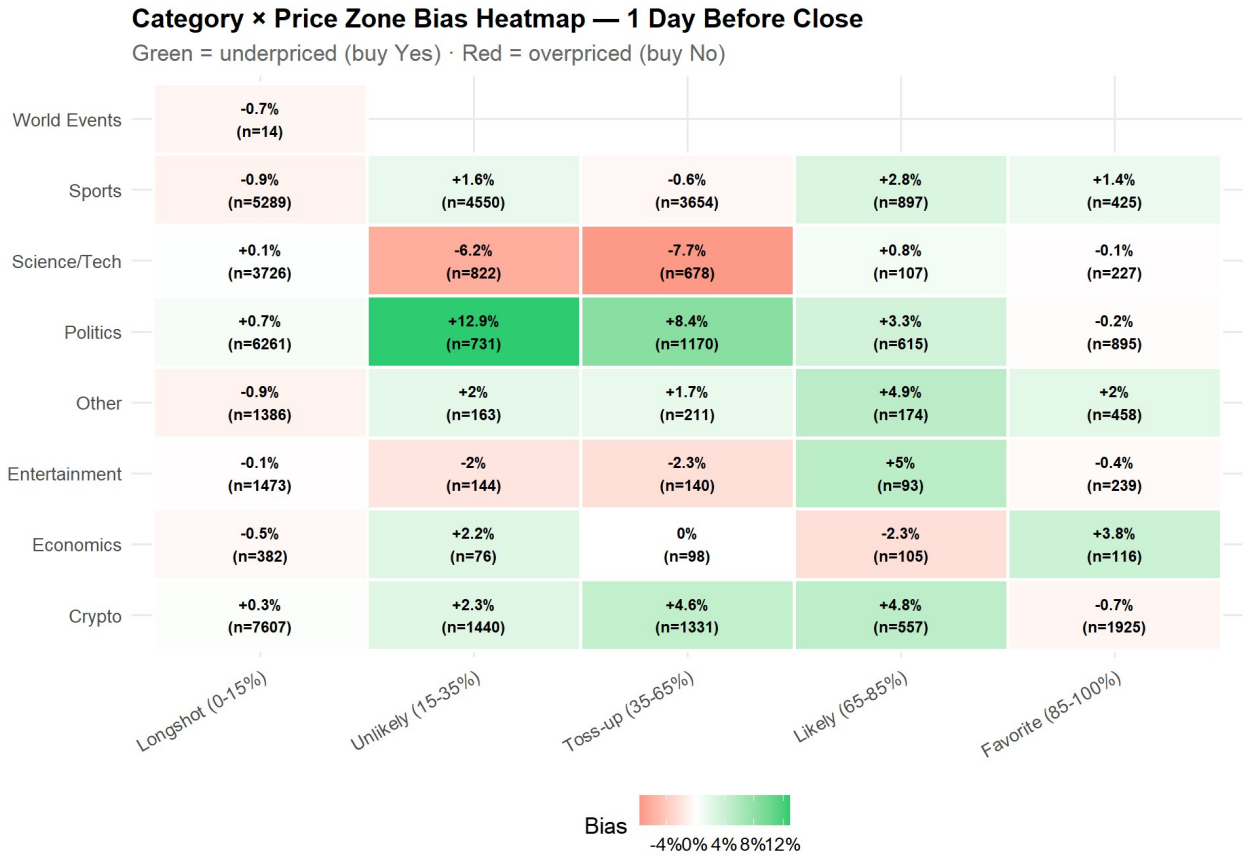
Show





3.2 Category Bias Heatmap

Show



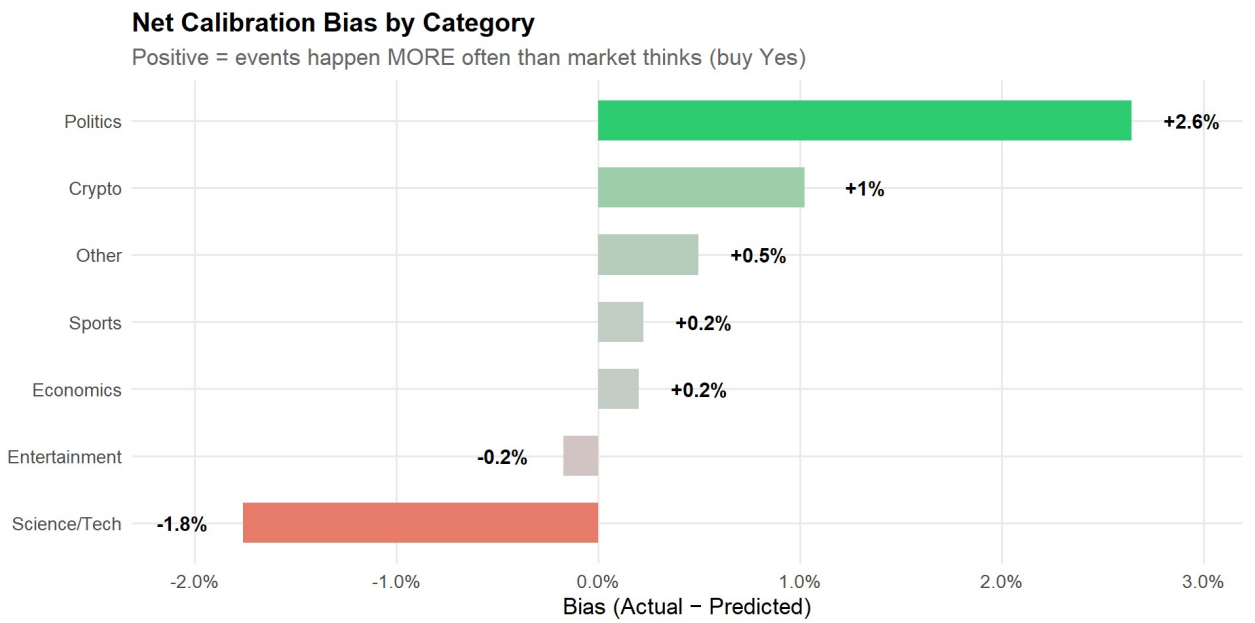
3.3 Aggregate Category Bias & Brier Score

Show

Category-Level Bias & Accuracy (1 day before close)

category	n_markets	avg_price	resolution_rate	bias	brier_score	abs_bias
World Events	18	0.1137	0.2222	0.1085	0.0650	0.1193
Politics	9672	0.2265	0.2529	0.0264	0.0755	0.1431
Science/Tech	5560	0.1656	0.1480	-0.0176	0.0743	0.1533
Crypto	12860	0.2694	0.2796	0.0102	0.0732	0.1438
Other	2392	0.3098	0.3148	0.0050	0.0554	0.1200
Sports	14815	0.2823	0.2846	0.0022	0.1407	0.2826
Economics	777	0.3403	0.3423	0.0020	0.0846	0.1735
Entertainment	2089	0.2061	0.2044	-0.0017	0.0473	0.0972

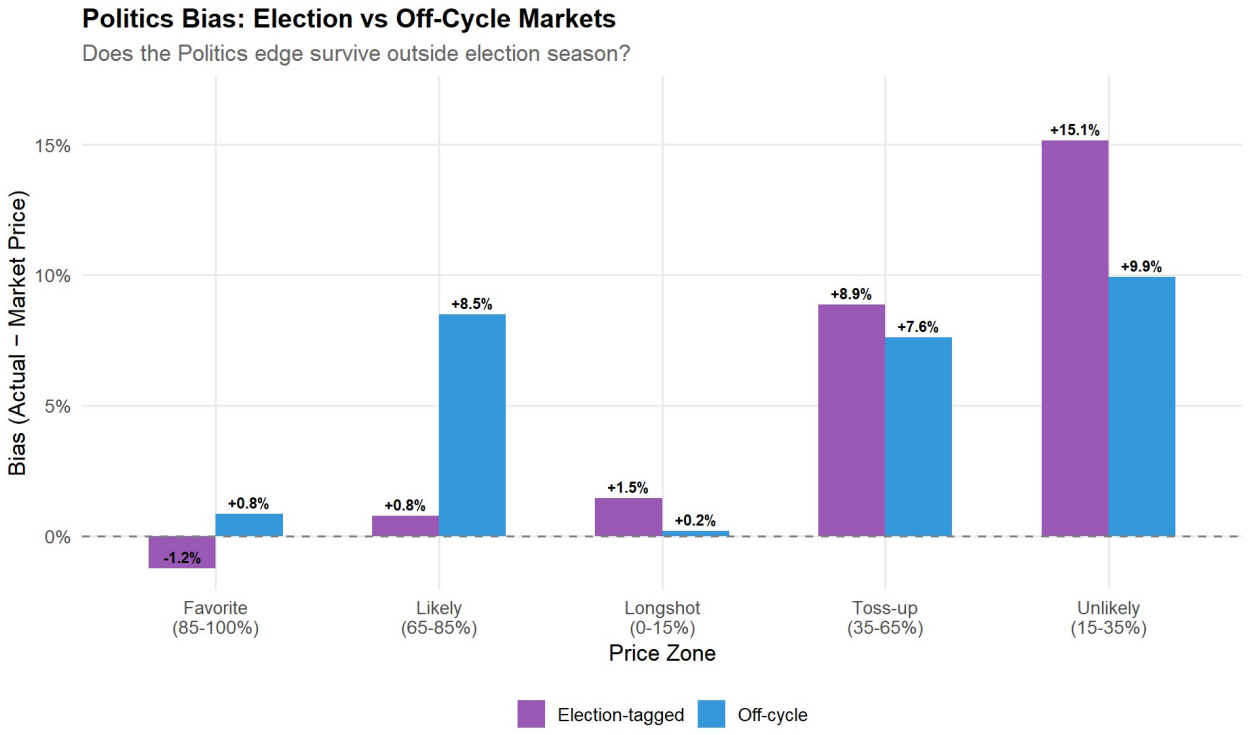
Show



3.4 Politics: Election vs. Non-Election Markets

New. The Politics category shows the strongest aggregate bias (+2.6%). We check how much of this is 2024 US election spillover — if the edge is election-specific, it may not persist in 2026.

Show



Show

Politics: Election vs Off-Cycle Bias (1 day before close)

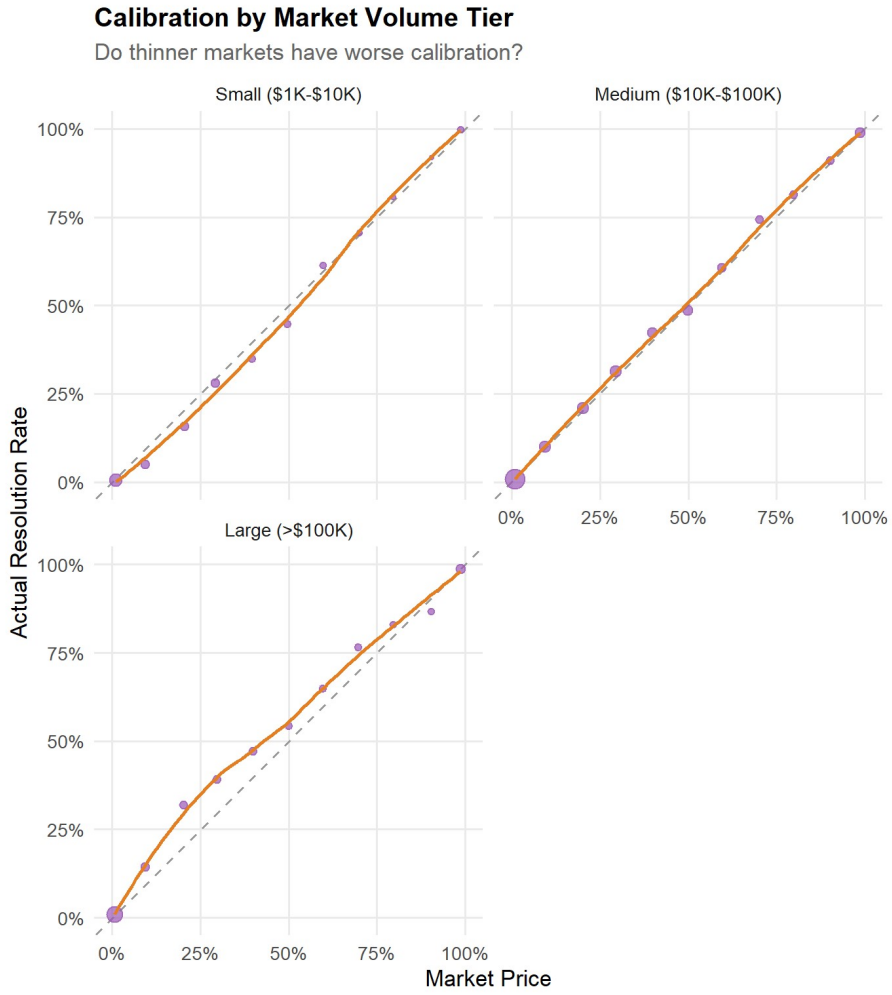
market_type	n_markets	avg_price	resolution_rate	bias	brier_score
Election-tagged	4476	0.2844	0.3213	0.0369	0.1054

market_type	n_markets	avg_price	resolution_rate	bias	brier_score
Off-cycle	5196	0.1766	0.1940	0.0174	0.0497

4 · Volume-Weighted Analysis

Thin markets are noisier. Let's check if bias patterns change when we weight by volume.

Show



Show

Brier Score & Bias by Volume Tier

vol_tier	n	brier_score	bias
Small ($1K-$10K)	8650	0.0917	-0.0188
Medium ($10K-$100K)	26440	0.0958	0.0058
Large (>$100K)	13093	0.0872	0.0263

5 · Post-Breakout Over/Under Correction

New. A "breakout" is a large price move (≥10pp) within 24 hours. After such a move, does the price continue in the same direction (momentum) or reverse (overcorrection)? This is directly

exploitable by a rule-based trading bot.

Show

```
## 40351 markets qualify for breakout analysis
```

Show

```
##            used  (Mb) gc trigger   (Mb)   max used    (Mb)
## Ncells  4940142 263.9    9375825  500.8    9375825   500.8
## Vcells 63906941 487.6 1142691488 8718.1 1428218670 10896.5
```

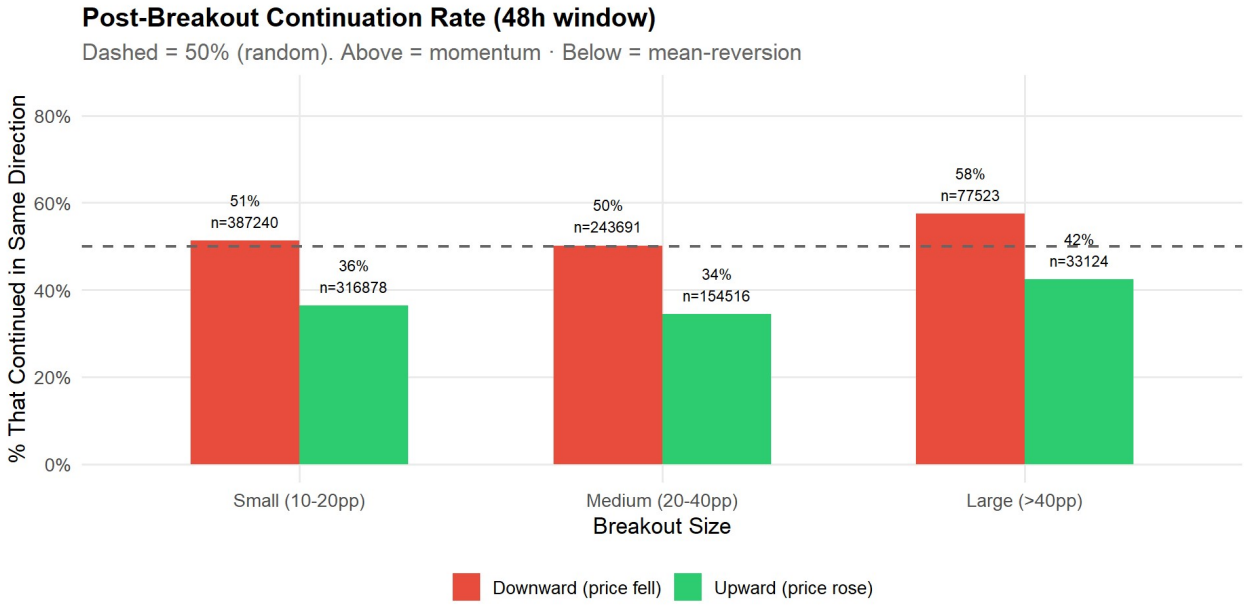
Show

```
## 1212972 breakout events detected
```

Show

```
## Unique markets with ≥1 breakout: 33591
```

Show

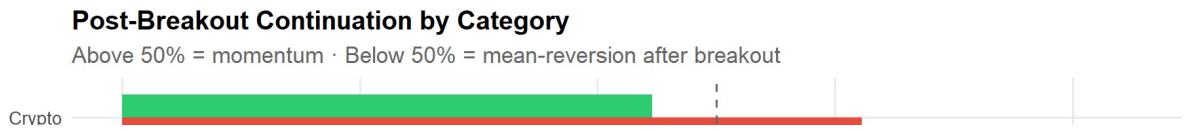


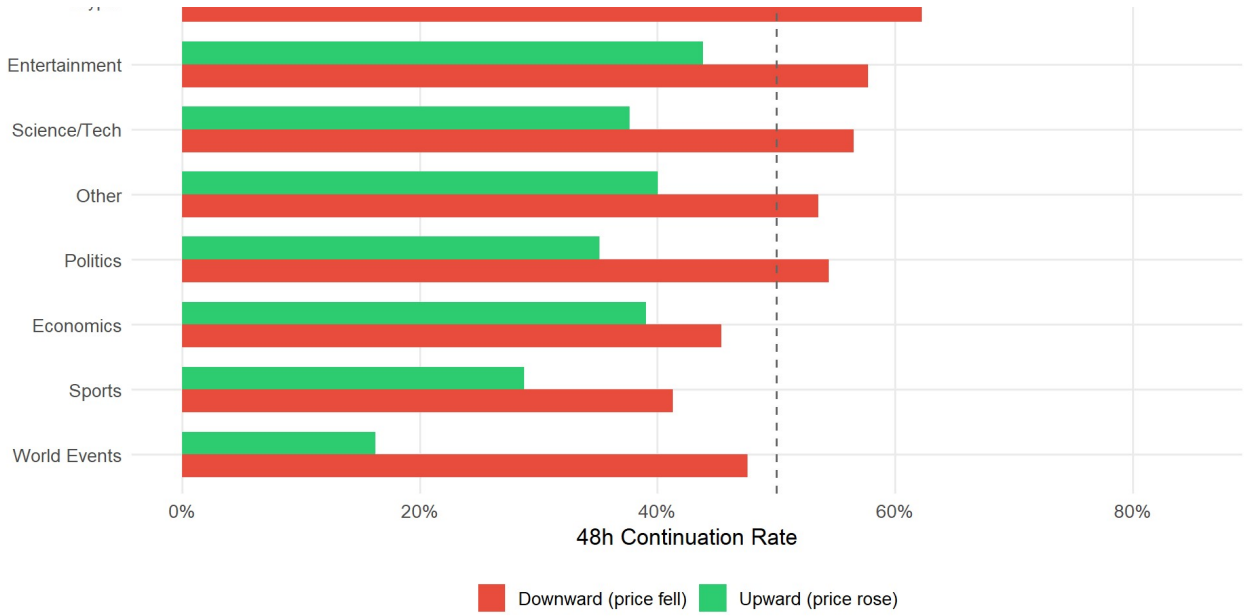
Show

Breakout Direction vs Eventual Resolution — Are Breakouts Informative?

breakout_dir	n	correct_dir_pct	avg_post_48h
Downward (price fell)	708454	0.7792	0.0099
Upward (price rose)	504518	0.4349	-0.0578

Show





6 · Market Duration & Age Bias

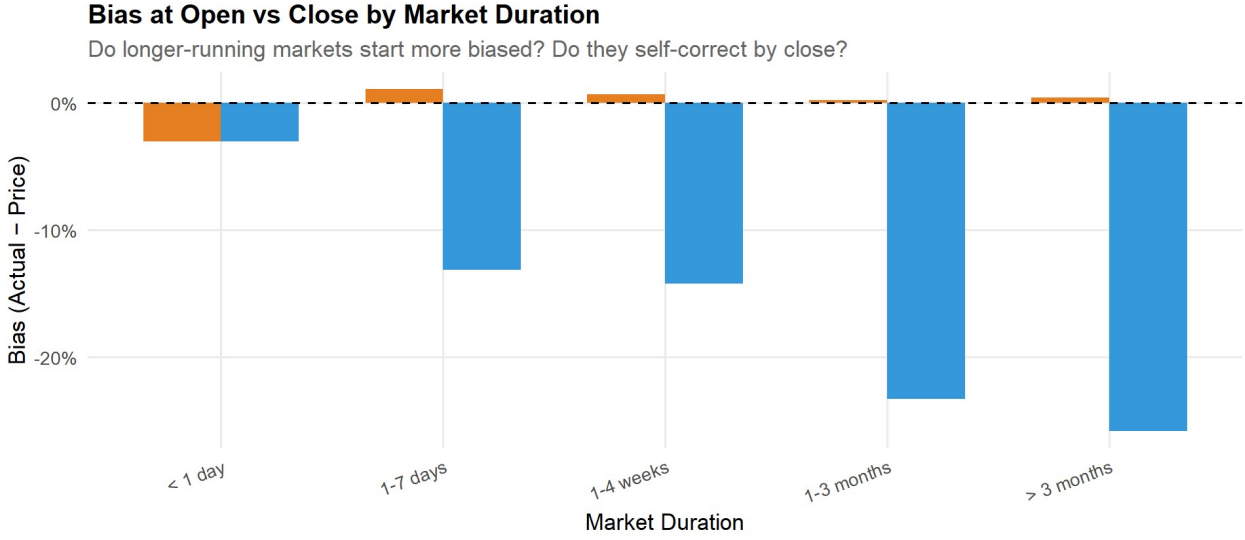
New. Markets running for 1 week vs 6 months calibrate very differently. Longer markets attract more uncertainty at open; shorter ones may be efficiently priced from the start.

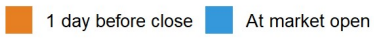
Show

Bias & Calibration by Market Duration

duration_tier	n	init_bias	final_bias	price_range	resolution_r	brier_initial
< 1 day	675	-0.0300	-0.0300	0.4651	0.4089	0.1925
1-7 days	15075	-0.1312	0.0109	0.5112	0.2825	0.2036
1-4 weeks	24046	-0.1422	0.0068	0.5304	0.2810	0.2057
1-3 months	4915	-0.2327	0.0025	0.4973	0.1727	0.1928
> 3 months	3472	-0.2581	0.0044	0.4753	0.1129	0.1776

Show



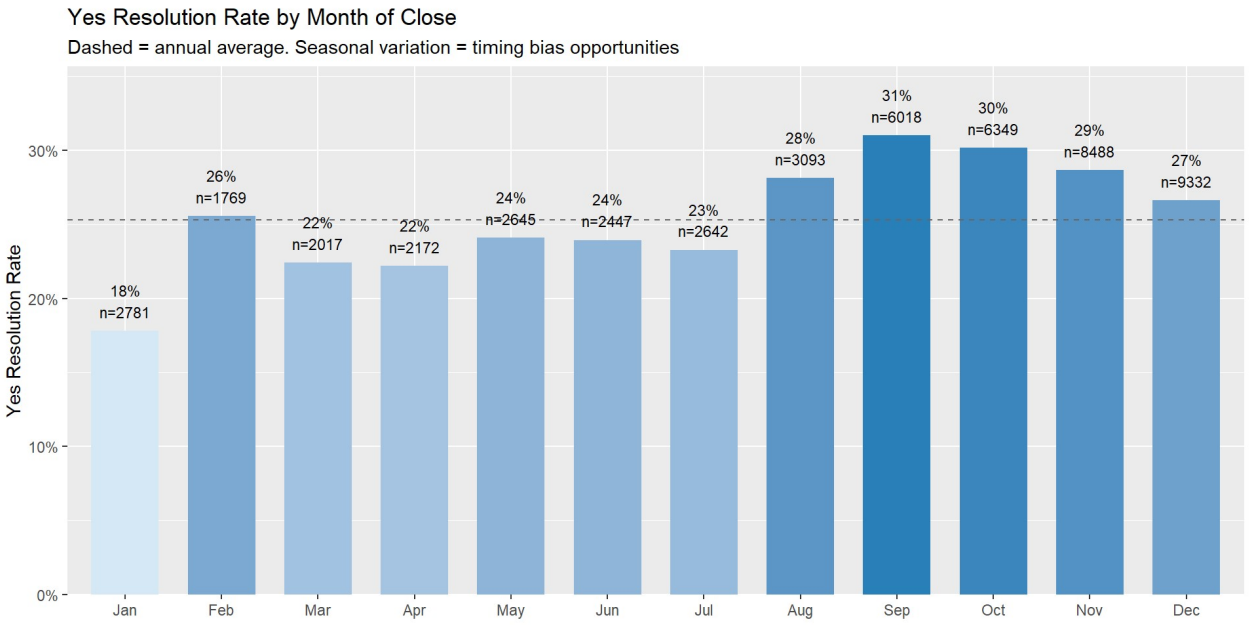


7 · Temporal & Resolution Timing Patterns

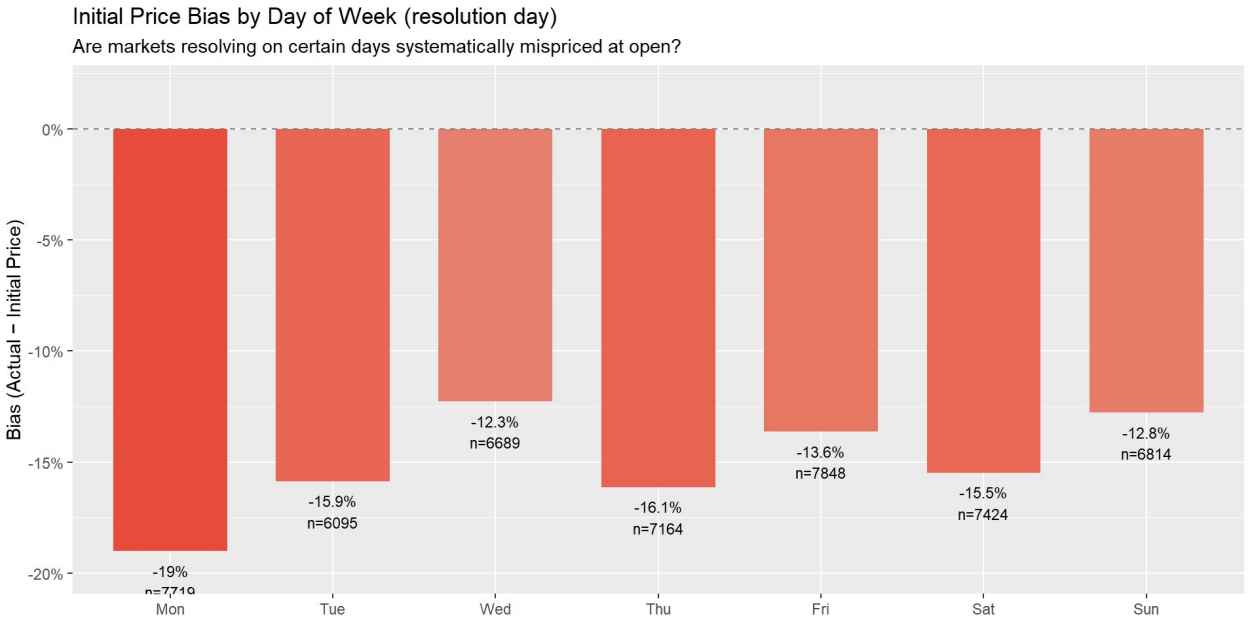
New. Systematic biases based on *when* markets resolve. Day-of-week and monthly seasonality can generate exploitable patterns — sports markets resolve heavily on weekends, politics on election days, etc.

Show

Show



Show



Show

Calibration by Quarter — Is the market getting more efficient over time?

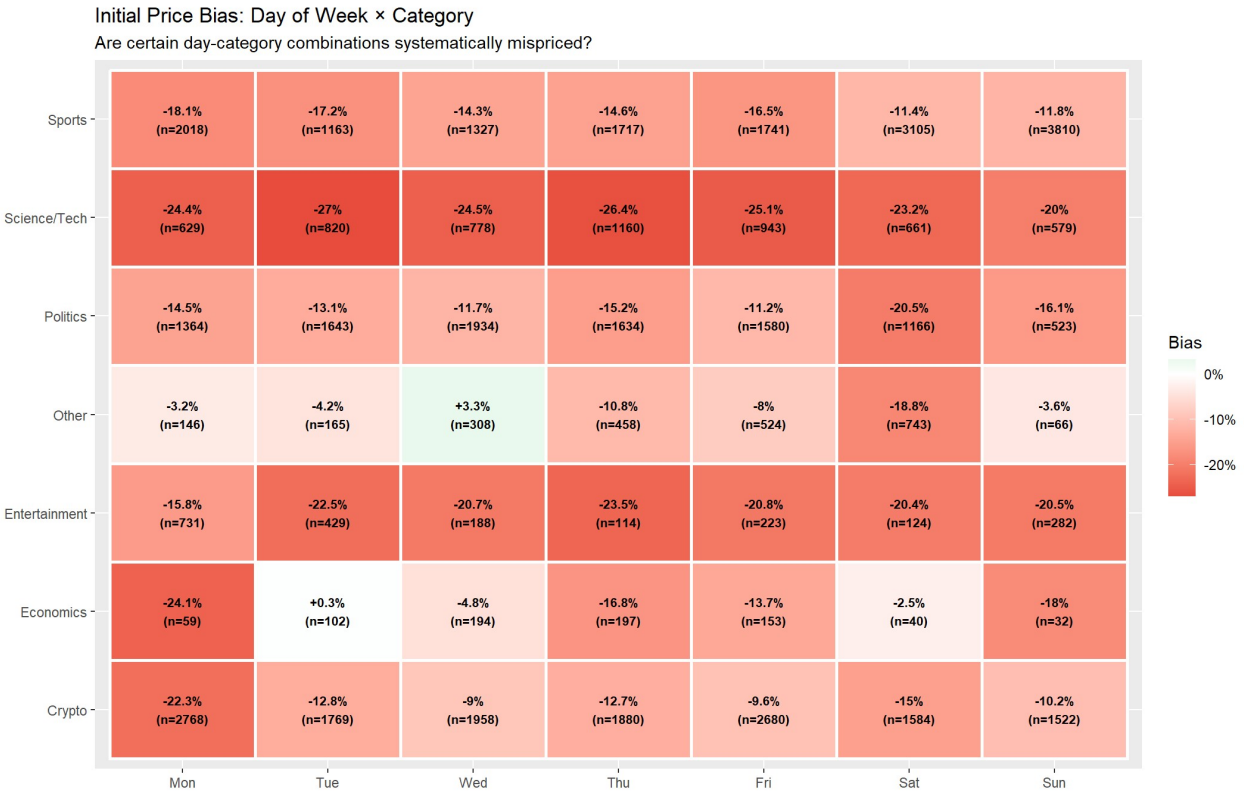
quarter	n	bias	brier_score	abs_bias

quarter	n	bias	brier_score	abs_bias
Q1	6474	-0.0006	0.0736	0.1466
Q2	7213	0.0070	0.0890	0.1792
Q3	10443	0.0078	0.0965	0.1922
Q4	24053	0.0086	0.0974	0.1933

7.5 · Day-of-Week Bias by Category

New. Different categories resolve on different days. Sports on weekends, politics on weekdays. Do mispricing patterns differ by day × category?

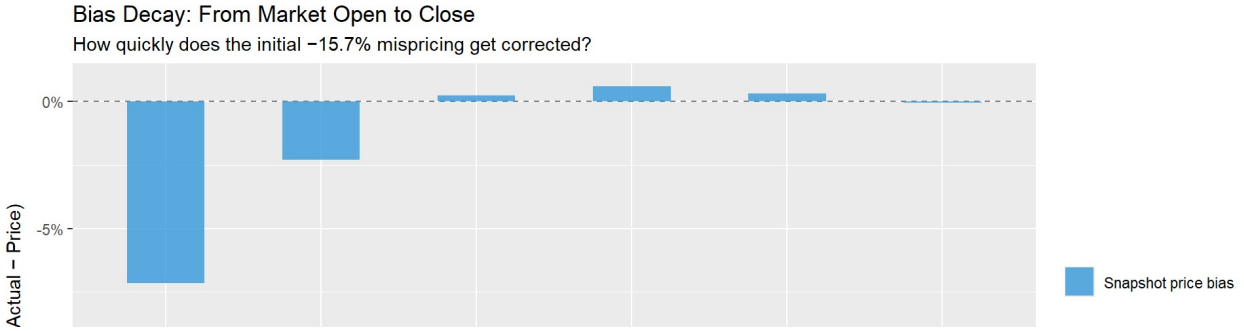
Show

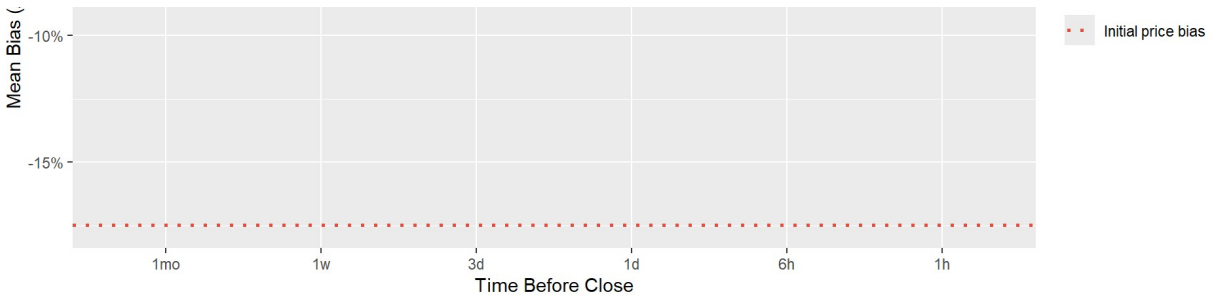


7.6 · New Market Opening Window

New. The initial price bias of −15.7% suggests a systematic "buy Yes on new markets" edge. We test how quickly this corrects — is the best window the first hour, first day, or first week?

Show

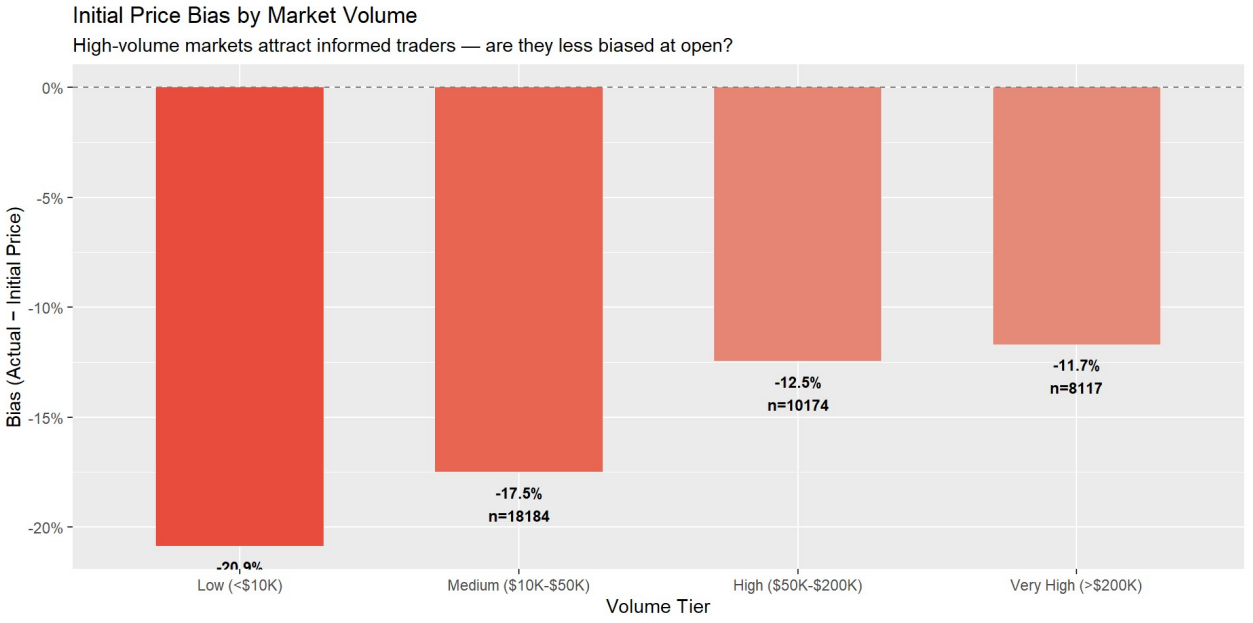




7.7 · Volume Surge as Directional Signal

New. Do sudden volume spikes predict resolution direction? If informed traders enter aggressively, volume surges might precede the correct outcome.

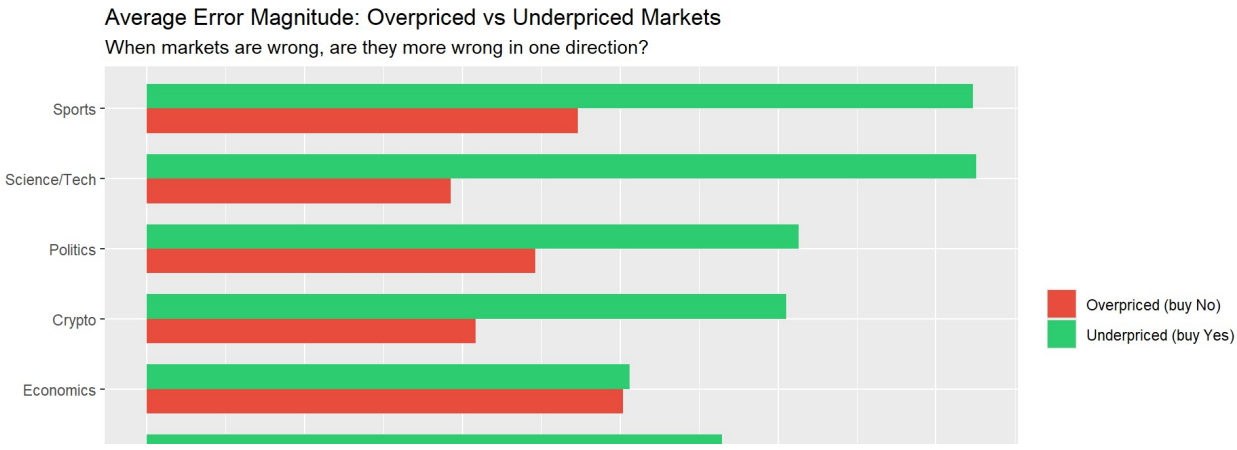
Show

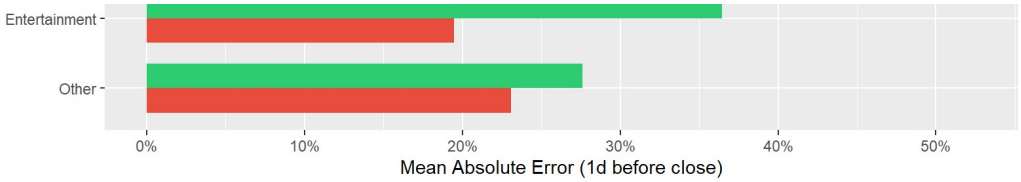


7.8 · Asymmetric Error: Over- vs Under-priced Markets

New. Do markets make bigger errors when overpricing Yes (price > resolution) vs underpricing Yes? If the error is asymmetric, it tells us which side of the trade is systematically more profitable.

Show





Show

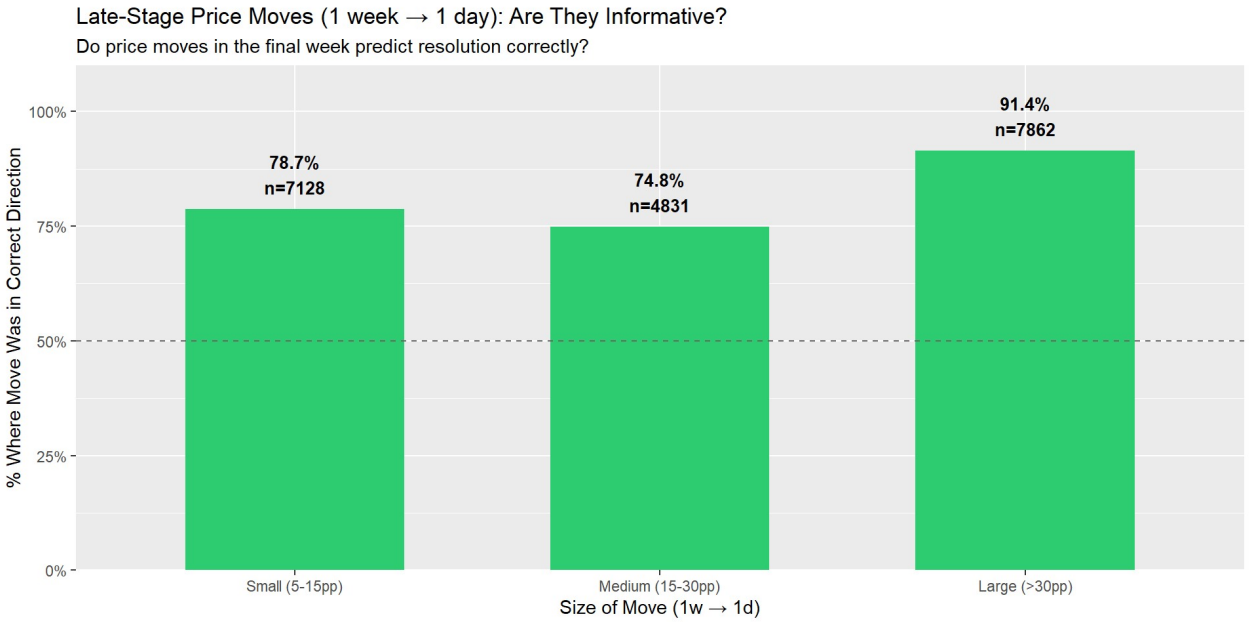
Asymmetric Error Distribution (1d before close)

error_dir	n	pct	avg_error	avg_profit
Approximately correct	20562	0.4267	0.0048	0.0000
Overpriced (buy No)	17445	0.3621	0.2406	0.2406
Underpriced (buy Yes)	10176	0.2112	0.4521	0.4521

7.9 · Spread Between Correlated Snapshots

New. If the 1-week price and 1-day price for the same market diverge substantially, it signals either new information or a slow-moving correction. Markets where the 1w→1d price *increased* but resolution was 0 (or vice versa) may indicate exploitable late-stage mispricing.

Show



8 · Exploitable Pattern Summary (with Bootstrap CIs)

Show

Top 20 Most Exploitable Category × Price Zone Combinations (|bias| > 3%, n ≥ 15)

category	price_zone	n_markets	avg_price	actual_rate	bias	edge_pct	brier

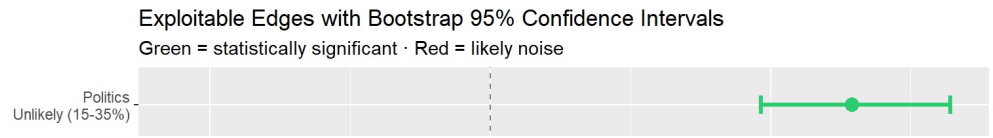
category	price_zone	n_markets	avg_price	actual_rate	bias	edge_pct	brier
Politics	Unlikely (15-35%)	731	0.253	0.382	0.129	51.071	0.247
Politics	Toss-up (35-65%)	1170	0.502	0.586	0.084	16.802	0.247
Science/Tech	Toss-up (35-65%)	678	0.465	0.388	-0.077	-16.506	0.232
Science/Tech	Unlikely (15-35%)	822	0.251	0.189	-0.062	-24.751	0.155
Entertainment	Likely (65-85%)	93	0.756	0.806	0.050	6.656	0.152
Other	Likely (65-85%)	174	0.761	0.810	0.049	6.471	0.156
Crypto	Likely (65-85%)	557	0.751	0.799	0.048	6.371	0.164
Crypto	Toss-up (35-65%)	1331	0.482	0.527	0.046	9.444	0.243
Economics	Favorite (85-100%)	116	0.945	0.983	0.038	3.976	0.019
Politics	Likely (65-85%)	615	0.749	0.782	0.033	4.470	0.169

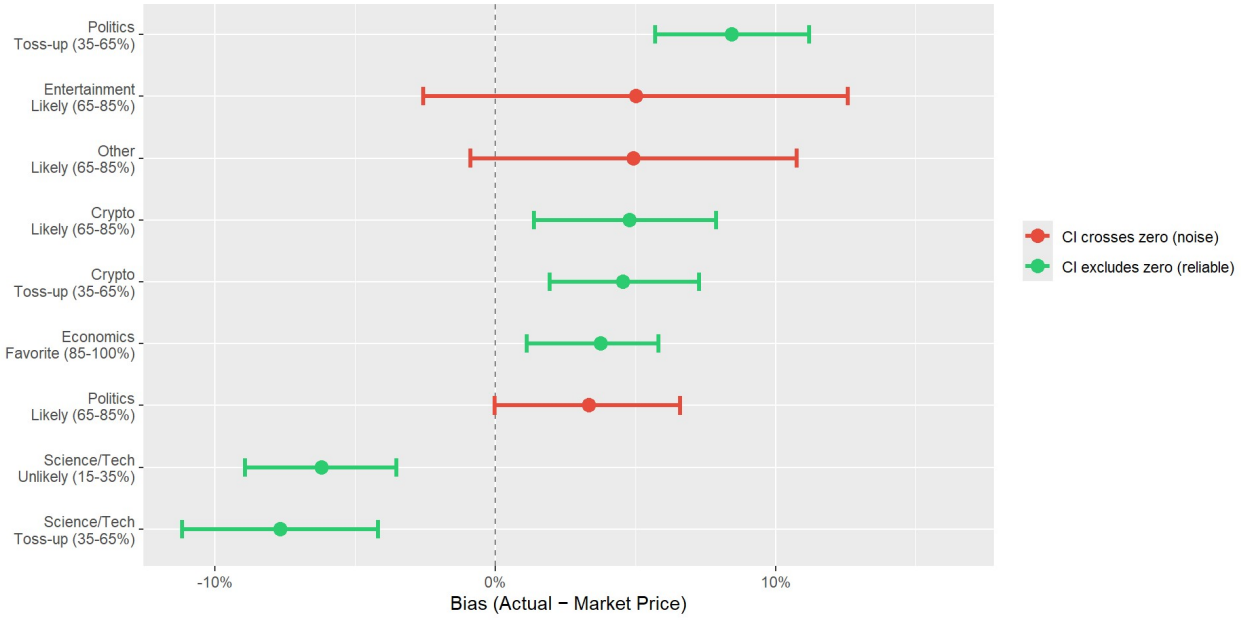
Show

Bootstrap 95% CIs — Which Edges Are Statistically Real?

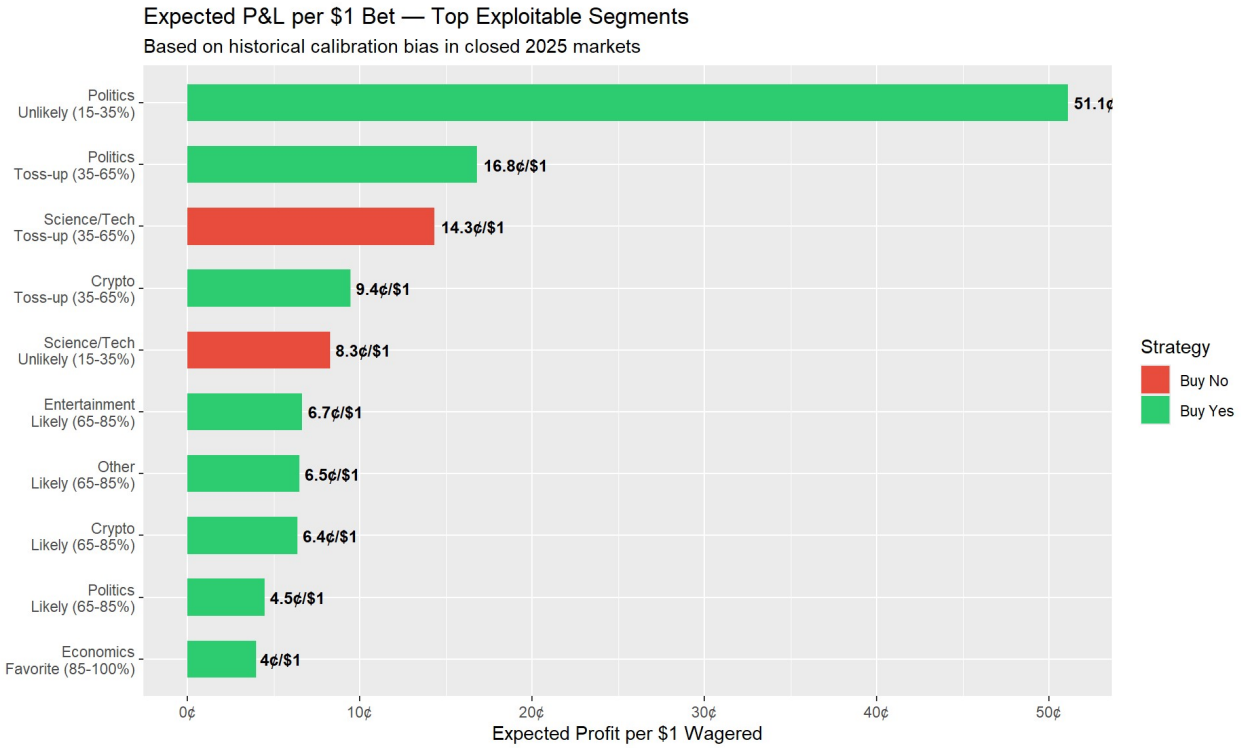
category	price_zone	n_obs	point_bias	ci_lo	ci_hi	significant
Politics	Unlikely (15-35%)	731	12.9%	9.67%	16.41%	YES
Politics	Toss-up (35-65%)	1170	8.43%	5.7%	11.21%	YES
Science/Tech	Toss-up (35-65%)	678	-7.67%	-11.17%	-4.19%	YES
Science/Tech	Unlikely (15-35%)	822	-6.2%	-8.94%	-3.53%	YES
Entertainment	Likely (65-85%)	93	5.03%	-2.56%	12.59%	no
Other	Likely (65-85%)	174	4.93%	-0.89%	10.76%	no
Crypto	Likely (65-85%)	557	4.79%	1.39%	7.88%	YES
Crypto	Toss-up (35-65%)	1331	4.55%	1.95%	7.27%	YES
Economics	Favorite (85-100%)	116	3.76%	1.13%	5.82%	YES
Politics	Likely (65-85%)	615	3.35%	-0.01%	6.6%	no

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9 · Key Takeaways

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Automated Findings

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- **Overall market bias** (1d before close): +0.7%. Markets are approximately well-calibrated overall.

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- **Initial price bias** (at market open): -15.7%. This is **larger** than 1-day bias — the best

mispricing window is right after market creation.

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- **Most biased category**: World Events (+10.8% bias, n=18)

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- **Best calibrated category**: Entertainment (Brier score: 0.0473)

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- **Largest exploitable edge**: Politics × Unlikely (15-35%) (+12.9% bias, n=731)

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- **Post-breakout behaviour**: 45% of breakouts continue in the same direction over 48h. No clear momentum or reversion signal.

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- **Statistically reliable edges**: 7/10 top segments have bootstrap CIs excluding zero.

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> **Disclaimer**: Past calibration patterns do not guarantee future exploitability. Transaction costs, slippage, and market evolution may erode these edges. Always paper-trade or use minimal size before scaling.

Analysis generated on 2026-02-18 using 49753 closed binary markets and price history data.